Exercise 6.2 SOLUTION

1. **Find the common difference and write the next two terms of each arithmetic sequence.**

i) $9,16,23,\ldots$

ii) $5,5+\sqrt{2},5+2\sqrt{2},\ldots$

Sol:

i) $9,16,23,\ldots$

Common difference: $d=16-9=7$

Next two terms:

$23+7=30$

$30+7=37$

ii) $5,5+\sqrt{2},5+2\sqrt{2},\ldots$

Common difference: $d=\cancel{5}+\sqrt{2}-\cancel{5}=\sqrt{2}$

Next two terms:

$5+2\sqrt{2}+\sqrt{2}=5+3\sqrt{2}$

$5+3\sqrt{2}+\sqrt{2}=5+4\sqrt{2}$

2. **Write the first three terms of each arithmetic sequence, with given information.**

i) $a_1=2,d=13$ ii) $a_1=12,d=-13$

Sol:

i) $a_1=2,d=13$

First term $(a_1)=2$

Second term $a_2=a_1+d=2+13=15$

Third term $(a_3)=a_2+d=15+13=28$

The first three terms are $2,15,28$.

ii) $a_1=12,d=-13$

First term $(a_1)=12$

Second term $(a_2)=a_1+d=12+(-13)=-1$

Third term $a_3=a_2+d=-1+(-13)=-1-13=-14$

The first three terms are $12,-1,-14$

3. **Find a_{n+1} and a_{2n} if $a_n=4+3n$**

Sol: $a_n=4+3n$

We need to find a_{n+1} and a_{2n}.

To find a_{n+1}, replace $n+1$ by n in the formula for a_n.

$a_{n+1}=4+3(n+1)$

$a_{n+1}=4+3n+3$

$a_{n+1}=3n+7$

To find a_{2n}, replace $2n$ by n in the formula for a_n:

$a_{2n}=4+3(2n)$

$a_{2n}=4+6n$

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4. **Find the indicated term of each of the following arithmetic sequences:**

i) $a_1 = 3, d = 7, a_{14}$, **ii)** $8, 3, -2, \ldots, a_{12}$

Sol:

i) $a_1 = 3, d = 7$, find a_{14}

Here, $a_1 = 3, d = 7$, and $n = 14$.

Substitute these values into the formula: $a_n = a_1 + (n-1)d$

$a_{14} = 3 + (14-1)7 = 3 + (13)7$

$a_{14} = 3 + 91 = 94$

ii) $8, 3, -2, \ldots$, find a_{12}

Here, $a_1 = 8, d = 3 - 8 = -5$, and $n = 12$

Substitute these values into the formula: $a_n = a_1 + (n-1)d$

$a_{12} = 8 + (12-1)(-5)$

$a_{12} = 8 + (11)(-5) = 8 - 55 = -47$

5. **The 18th and 30th terms of an arithmetic sequence are 367 and 499 respectively. How many terms of this sequence are less than 1000?**

Sol: Given: $a_{18} = 367$ and $a_{30} = 499$

The formula for the nth term of an arithmetic sequence is $a_n = a_1 + (n-1)d$.

$a_{18} = a_1 + (18-1)d \Rightarrow 367 = a_1 + 17d$ ---------(1)

$a_{30} = a_1 + (30-1)d \Rightarrow 499 = a_1 + 29d$ ---------(2)

Subtract (1) and (2):

$(a_1 + 29d) - (a_1 + 17d) = 499 - 367$

$12d = 132$

$d = 11$

Substitute $d = 11$ into equation (1):

$367 = a_1 + 17(11) \Rightarrow 367 = a_1 + 187$

$a_1 = 367 - 187 \Rightarrow a_1 = 180$

$a_n = a_1 + (n-1)d$

$a_n = 180 + (n-1)11$

We want to find how many terms of this sequence are less than 1000.

$180 + (n-1)11 < 1000 \Rightarrow (n-1) < \dfrac{1000 - 180}{11}$

$n - 1 < \dfrac{820}{11}$

$n - 1 < 74.545\ldots$

Add 1 to both sides:

$n < 74.545\ldots + 1$

$n < 75.545\ldots$

Since n must be an integer (the term number), the largest integer value of n that is less than $75.545\ldots$ is 75.

Therefore, there are 75 terms of this sequence that are less than 1000.

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6. **Is 301 a term the A.P 5,11,17,.....?**

Sol: Given A.P.: $5,11,17,...$

Here

$a_1 = 5,\ d = 11-5 = 6$

Use the formula for the nth term of an A.P.:

$a_n = a_1 + (n-1)d$

Take $a_n = 301$ and solve for n:

$301 = 5 + (n-1)6$

$301 - 5 = (n-1)6$

$296 = (n-1)6$

$\frac{296}{6} = (n-1)$

$\frac{148}{3} = (n-1)$

$49.33... = n-1$

$n = 50.33...$

Since n is not an integer, 301 is not a term of the A.P. $5,11,17,...$

7. **If $2x, x+8, 3x+1$ are in A.P., then find the value of x.**

Sol: Given that $2x, x+8$ and $3x+1$ are in A.P., we use the property:

$3x+1-(x+8) = x+8-2x$

$3x+1-x-8 = 8-x \Rightarrow 2x-7 = 8-x$

$\Rightarrow 2x+x = 15 \Rightarrow 3x = 15 \Rightarrow x = 5$

8. **Which term of the A.P., $3,8,13,...$ is 123?**

Sol: Given that $3,8,13,...$

Here, $a_1 = 3, d = 8-3 = 5, a_n = 123.$

Using the formula for the nth term of an A.P.:

$a_n = a_1 + (n-1)d$

$123 = 3 + (n-1)5$

$123 - 3 = (n-1)5$

$120 = 5(n-1)$

$\frac{120}{5} = n-1$

$24 = n-1$

$n = 24+1$

$n = 25$

123 is the 25th term of the sequence.

9. **Which term of the A.P., $30, 29.5, 29,...$ is the first negative term?**

Sol: Given A.P.: $30, 29.5, 29,...$

Here, $a_1 = 30$ and $d = 29.5 - 30 = -0.5.$

We need to find the first term a_n such that $a_n < 0$.

The formula for the nth term of an A.P. is

$a_n = a_1 + (n-1)d$

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Set $a_n < 0$:

$30+(n-1)(-0.5)<0$

$30-0.5(n-1)<0$

$30<0.5(n-1)$

$\frac{30}{0.5}<n-1 \Rightarrow 60<n-1 \Rightarrow 60+1<n \Rightarrow 61<n$

Since n must be an integer, the smallest integer value for n satisfying $n>61$ *is* 62.

The first negative term is the 62nd term.

10. The 7th and 21st term of an A.P., are 37 and 107 respectively. Find the A.P. and its 100th term.

Sol: Given $a_7 = 37$ and $a_{21} = 107$

By using $a_n = a_1 + (n-1)d$

$a_7 = 37 \Rightarrow a_1 + 6d = 37 \quad (1)$

$a_{21} = 107 \Rightarrow a_1 + 20d = 107 \quad (2)$

Subtract (1) from (2):

$(a_1 + 20d) - (a_1 + 6d) = 107 - 37$

$14d = 70$

$d = \frac{70}{14} \Rightarrow d = 5$

Substitute $d = 5$ into (1):

$a_1 + 6(5) = 37 \Rightarrow a_1 + 30 = 37$

$a_1 = 37 - 30 \Rightarrow a_1 = 7$

The A.P. is $a_1, a_1 + d, a_1 + 2d, \ldots$

$A.P. = 7, 7+5, 7+2(5), \ldots.$

A.P is $7, 12, 17, \ldots.$

Now, find the 100th term (a_{100}):

$a_{100} = a + (100-1)d = 7 + (99)5$

$a_{100} = 7 + 495 = 502$

The 100th term is 502.

11. If $\frac{1}{a-c}, \frac{1}{b-c}, \frac{1}{b-a}$ are in A.P.,

The show that $\frac{a-b}{a-c} = \frac{a-c}{b-a}$.

Sol: Given $\frac{1}{a-c}, \frac{1}{b-c}$ and $\frac{1}{b-a}$ are in AP then

$\frac{1}{b-c} - \frac{1}{a-c} = \frac{1}{b-a} - \frac{1}{b-c}$

$\frac{a-c-(b-c)}{(b-c)(a-c)} = \frac{b-c-(b-a)}{(b-a)(b-c)}$

$\frac{a-b}{(b-c)(a-c)} = \frac{a-c}{(b-a)(b-c)}$

$\frac{a-b}{a-c} = \frac{a-c}{b-a}$

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12. **How many number of three digits are divisible by 7?**

Sol: There digit numbers divisible by 7 between 100 and 999 are 105, 112, 119,..., 994

$a_1 = 105, d = 7, a_n = 994, n = ?$

$a_n = a_1 + (n-1)d$

$994 = 105 + (n-1)7 \Rightarrow 994 - 105 = (n-1)7$

$\Rightarrow (n-1) = \frac{889}{7} \Rightarrow n - 1 = 127 \Rightarrow n = 128$

Hence 128 three digits numbers are divisible by 7 between 100 and 999.

13. **Find the 8th term from the end of the A.P., $8, 11, 14, ..., 185$**

Sol': To find the 8th term from the end of the A.P., $8, 11, 14, ..., 185$:

Here $d = 11 - 8 = 3$, $a_1 = 8, a_n = 185$ from start

$185 = 8 + (n-1)3$

$177 = (n-1)3 \Rightarrow 59 = n - 1$

$n = 60$

Total terms are 60 so 8th term from the end is 53th term.

$a_{53} = 8 + (53-1)3$

$a_{53} = 8 + (52)3$

$a_{53} = 8 + 156 = 164$

Here 164 is eight term from the end.

14. **Find the nth term of progression $\left(\frac{3}{7}\right)^{10}, \left(\frac{10}{7}\right)^{10}, \left(\frac{17}{7}\right)^{10}$,....Is the progression an A.P?**

Sol: Given progression: $\left(\frac{3}{7}\right)^{10}, \left(\frac{10}{7}\right)^{10}, \left(\frac{17}{7}\right)^{10}, ...$

Consider the bases: $\frac{3}{7}, \frac{10}{7}, \frac{17}{7}, ...$

Here $a_1 = \frac{3}{7}, d = \frac{10}{7} - \frac{3}{7} = \frac{7}{7} = 1, n = n$

$a_n = a_1 + (n-1)d$

$a_n = \frac{3}{7} + (n-1)(1) = \frac{3}{7} + n - 1$

$= \frac{3 + 7n - 7}{7} = \frac{7n - 4}{7}$

Hence nth term $= \left(\frac{7n-4}{7}\right)^{10}$

To check given progression is an A.P.

d (Difference of 1st & 2nd) $= \left(\frac{10}{7}\right)^{10} - \left(\frac{3}{7}\right)^{10} = \frac{10^{10} - 3^{10}}{7^{10}}$

d (Difference of 2nd & 3rd) $= \left(\frac{17}{7}\right)^{10} - \left(\frac{10}{7}\right)^{10} = \frac{17^{10} - 10^{10}}{7^{10}}$

Clearly $\frac{10^7 - 3^{10}}{7^{10}} \neq \frac{17^{10} - 10^{10}}{7^{10}}$

Hence it is not an A.P.

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15. **If the arithmetic progressions $3,10,17,\ldots$ and $63,65,67,\ldots.$ are such that their nth terms are equal, then find the value of n.**

Sol: First A.P.: $3,10,17,\ldots$

$a_1 = 3,\ d_1 = 10-3 = 7,\ n = n$

$a_n = a_1 + (n-1)d = 3+(n-1)7$

Second A.P.: $63,65,67,\ldots.$

$a_2 = 63,\ d_2 = 65-63 = 2,\ n = n$

Nth term: $b_n = 63+(n-1)2$

Given $a_n = b_n$:

$3+(n-1)7 = 63+(n-1)2$

$3+7n-7 = 63+2n-2$

$7n-4 = 2n+61$

$7n-2n = 61+4 \Rightarrow 5n = 65 \Rightarrow n = 13$

The value of n is 13.

16. **If the pth term of an A.P. is q and the qth term is p, prove that its nth term is $(p+q-n)$.**

Sol: Given an A.P. where:

$a_p = q \Rightarrow a+(p-1)d = q \quad (1)$

$a_q = p \Rightarrow a+(q-1)d = p \quad (2)$

Subtract (2) from (1)

$(a+(p-1)d)-(a+(q-1)d) = q-p$

$pd-d-qd+d = q-p$

$d(p-q) = -(p-q)$

$d = -1$ (assuming $p \neq q$)

Substitute $d = -1$ into (1):

$a+(p-1)(-1) = q$

$a-p+1 = q \Rightarrow a = p+q-1$

$a_n = a+(n-1)d$

$a_n = (p+q-1)+(n-1)(-1)$

$a_n = p+q-1-n+1$

$a_n = p+q-n$ Hence proved.

17. **If $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ are in A.P., show that $b = \frac{2ac}{a+c}$.**

Sol: Given $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ are in A.P.

So, $\frac{1}{c}-\frac{1}{b} = \frac{1}{b}-\frac{1}{a}$

Or $\frac{1}{c}+\frac{1}{a} = \frac{1}{b}+\frac{1}{b}$

$\frac{a+c}{ac} = \frac{2}{b} \Rightarrow \frac{ac}{a+c} = \frac{b}{2}$

$\Rightarrow b = \frac{2ac}{a+c}$ Hence Proved.

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18. If $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ are in A.P., show that the common difference is $\frac{a-c}{2ac}$.

Sol: Given $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ are in A.P.

So, $d = \frac{1}{c} - \frac{1}{b}$ (3rd − 2nd) $\quad (I)$

$d = \frac{1}{b} - \frac{1}{a}$ (2nd − 1st) $\quad (II)$

$I + II \Rightarrow 2d = \frac{1}{c} - \frac{1}{a}$

$2d = \frac{a-c}{ac} \Rightarrow d = \frac{a-c}{2ac}$ *Hence Proved*

19. If a_k and a_m denotes two different terms of an A.P., show that its nth term is $a_k + (n-k)\left(\frac{a_k - a_m}{k-m}\right)$.

Sol: Given an A.P. with terms a_k and a_m.

We know the general form of the jth term:

$a_j = a + (j-1)d$

So, we have:

$a_k = a + (k-1)d \quad (1)$

$a_m = a + (m-1)d \quad (2)$

Subtracting (2) from (1)

$a_k - a_m = [a + (k-1)d] - [a + (m-1)d]$

$a_k - a_m = a + kd - d - a - md + d$

$a_k - a_m = (k-m)d$

Assuming $k \neq m$:

$d = \frac{a_k - a_m}{k-m}$

Now, consider the nth term, a_n, we can express a_n in relation to a_k:

$a_n = a_k + (n-k)d$

Substitute the expression for d

$a_n = a_k + (n-k)\left(\frac{a_k - a_m}{k-m}\right)$

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n کی جگہ k لگائیں گے تو a_k آجائے گا اور n کی جگہ m لگائیں گے تو a_m آجائے گا۔

20. If $a_1, a_2, a_3, \ldots a_n$ are positive and in A.P., prove that $\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \ldots + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}} = \frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$

Sol: We first calculate L.H.S

For L.H.S $= \frac{1}{\sqrt{a_i}+\sqrt{a_i+1}} = \frac{1}{\sqrt{a_i}+\sqrt{a_i+1}} \cdot \frac{\sqrt{a_i}-\sqrt{a_i+1}}{\sqrt{a_i}-\sqrt{a_i+1}} = \frac{\sqrt{a_i}-\sqrt{a_i+1}}{(a_i)-(a_i+1)}$

Since $a_i + 1 = a_i + d$, then $a_i - a_i - 1 = -d$

$= \frac{\sqrt{a_i}-\sqrt{a_i+1}}{-d} = \frac{\sqrt{a_i+1}-\sqrt{a_i}}{d}$

$L.H.S = \sum_{i=1}^{n-1} \frac{\sqrt{a_i+1}-\sqrt{a_i}}{d} = \frac{1}{d}\left[\left(\sqrt{a_2}-\sqrt{a_1}\right) + \left(\sqrt{a_3}-\sqrt{a_2}\right) + \ldots + \left(\sqrt{a_n}-\sqrt{a_{n-1}}\right)\right]$

$= \frac{1}{d}\left[\cancel{\sqrt{a_2}} - \sqrt{a_1} + \sqrt{a_3} - \cancel{\sqrt{a_2}} + \ldots + \sqrt{a_n} - \cancel{\sqrt{a_{n-1}}}\right]$

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$$= \frac{\sqrt{a_n} - \sqrt{a_1}}{d} \quad \text{---------(I)}$$

$$\text{R.H.S} = \frac{n-1}{\sqrt{a_1} + \sqrt{a_n}} = \frac{n-1}{\sqrt{a_1} + \sqrt{a_n}} \cdot \frac{\sqrt{a_n} - \sqrt{a_1}}{\sqrt{a_n} - \sqrt{a_1}} = \frac{(n-1)\left(\sqrt{a_n} - \sqrt{a_1}\right)}{a_n - a_1}$$

$$a_n = a_1 + (n-1)d,$$

$$a_n - a_1 = (n-1)d.$$

$$= \frac{(n-1)\left(\sqrt{a_n} - \sqrt{a_1}\right)}{(n-1)d} = \frac{\sqrt{a_n} - \sqrt{a_1}}{d} \quad \text{---------(II)}$$

From I & II L.H.S = R.H.S

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21. **If the roots of the equation $(b-c)x^2 + (c-a)x + (a-b) = 0$ are equal. Show that a, b, c are in A.P.**

Sol: Given quadratic equation: $(b-c)x^2 + (c-a)x + (a-b) = 0$

For equal roots, the discriminant must be zero.

$\Delta = B^2 - 4AC = 0$

Here, $A = (b-c), B = (c-a), C = (a-b).$

$$(c-a)^2 - 4(b-c)(a-b) = 0$$

$$c^2 - 2ac + a^2 - 4(ab - b^2 - ac + bc) = 0$$

$$c^2 - 2ac + a^2 - 4ab + 4b^2 + 4ac - 4bc = 0$$

$$c^2 - 2ac + a^2 - 4ab + 2ac - 4bc = 0$$

$$a^2 + 4b^2 + c^2 - 4ab + 2ac - 4bc = 0$$

Rearrange to form a perfect square:

$$a^2 + (-2b)^2 + c^2 + 2(a)(-2b) + 2(a)(c) + 2(-2b)(c) = 0$$

So, $(a - 2b + c)^2 = 0$

Taking the square root of both sides:

$a - 2b + c = 0$

$a + c = 2b$

$\Rightarrow a + c = b + b \Rightarrow c - b = b - a$

Hence a, b, c are in A.P.

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22. **If the sides of a right angled triangle are in A.P., find the ratio of its sides.**

Sol: Let sides be $x - d, x, x + d$

Hypotenuse: $x + d$

$$(x-d)^2 + x^2 = (x+d)^2$$

$$x^2 - 2xd + d^2 + x^2 = x^2 + 2xd + d^2$$

$$2x^2 - 2xd + d^2 = x^2 + 2xd + d^2$$

$$x^2 - 4xd = 0$$

$$x(x - 4d) = 0$$

Since $x \neq 0$ (side length):

$x - 4d = 0 \Rightarrow x = 4d$

Sides are:

$x - d = 4d - d = 3d$

$x = 4d$

$x + d = 4d + d = 4d$

Ratio of sides $3d : 4d : 5d = 3 : 4 : 5$

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23. If the nth term of a progression is a linear expression in n, then prove that this progression is an A.P.

Sol: Let the nth term be T_n

Given that $T_n = An + B$ for constant A, B

$$T_{n+1} = A(n+1) + B = An + A + B.$$

$$T_{n+1} - T_n = (An + A + B) - (An + B) = An + A + B - An - B = A.$$

Also $T_{n-1} = A(n-1) + B = A_n - A + B$

$$T_n - T_{n-1} = A_n + B - A_n + A - B = A$$

$$T_{n+1} - T_n = T_n - T_{n-1}$$

Hence progression is an A.P.

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